THE UNIVERSITY OF BRITISH COLUMBIA
Faculty of Forestry & Environmental Stewardship

Dear Dr. Öpik,

Please consider our paper, "Warmer and longer growing seasons increase growth for juvenile trees in a common garden" for publication as a full paper in *New Phytologist*.

Anthropogenic climate change affects the timing of many biological processes, producing cascading effects (Piao *et al.*, 2019). For temperate trees, warming temperatures lengthen their growing season, which should increase growth (Gao *et al.*, 2022). However, recent evidence showed that trees may not grow more with warmer seasons which could be detrimental to the critical temperate forest carbon sink (Dow *et al.*, 2022; Silvestro *et al.*, 2023; Matula *et al.*, 2023). Our 'Full paper' adds a rare combination of full growing season length data via vegetative phenology, and growth via tree rings to address the question: do juvenile trees grow more with longer and warmer seasons. The favourable context of our common garden allows us to answer this fundamental question under drought- and competition-limited conditions. Our paper adds important information to the question that will help us understand whether forests will regenerate and store carbon with future climate.

What hypotheses or questions does this work address? Do juvenile trees grow more with longer and warmer seasons? We address this oustanding question in a common garden using juvenile trees of four species sourced from four provenances. We test effects of four different growing season metrics on growth and determine whether provenance latitude changed growth.

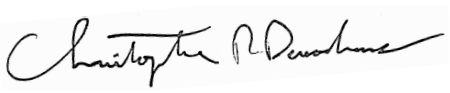
How does this work advance our current understanding of plant science? By a using multi-year, multi-species high quality phenology and growth dataset, we are able to tease apart the contribution of each season metric. Surprisingly, we show that end of season events—most often overlooked—appeared to be more important for growth than spring events.

Why is this work important and timely? Our work brings important insights on how longer, warmer seasons affect tree growth. Recent work has challenged the assumption that longer growing seasons increased tree growth, with major implications for the future of temperate forest carbon sink in a warming climate (Dow *et al.*, 2022; Silvestro *et al.*, 2023; Matula *et al.*, 2023).

All authors contributed to this work and approved this version for submission. The manuscript is 3977 words with a 282-word summary, and 4 figures and is not under consideration elsewhere. We hope you find it suitable for publication in *New Phytologist*, and look forward to hearing from you.

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Sincerely,



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References

Dow C, Kim AY, D'Orangeville L, *et al.* (2022) Warm springs alter timing but not total growth of temperate deciduous trees. *Nature*, **608**, 552–557. doi:10.1038/s41586-022-05092-3. URL `https://www.nature.com/articles/s41586-022-05092-3`.

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